



— • *Modular ERP for SMEs* • —

Project Proposal

for the Second year Group Project

SmartERP



by

Group IS24



Details Of the Supervisor and the Co-Supervisor

Proposed Project Supervisor (Academic Staff of UCSC):

Name of the supervisor: **Dr. Tharindu Wijethilake**

Signature of the supervisor:

Date:

Proposed Project Co-Supervisor (Assigned by Course Coordinator):

Name of the co-supervisor: **Ms. T. U. S. Wijewardhana**

Signature of the co-supervisor:

Date:

Table of Contents

1. Introduction to Project	6
Problem Statement.....	6
Proposed Solution.....	6
2. Project Goal	6
3. Scope of the Project	7
3.1 In Scope	7
1. Authentication and Role Management	7
2. Point of Sale (POS) and Invoice Management	7
3. Inventory and Purchase Management.....	7
4. Customer and Supplier Management	7
5. Employee and Task Management.....	8
6. Expense Management.....	8
7. Reporting and Business Insights	8
8. Domain-Specific Extensions	8
3.2 Out of Scope	9
1. Mobile Application Development	9
2. Advanced Financial Management	9
3. Third-Party System Integrations.....	9
4. E-Commerce and Online Services.....	9
5. Advanced Technologies	9
6. Additional Domain Extensions.....	9
4. Objectives of the Project.....	9
5. Project Feasibility	10
5.1 Technical Feasibility	10
5.2 Schedule Feasibility.....	10
5.3 Resource Feasibility	10
5.4 Economic Feasibility	10
5.5 Ethical and Legal Feasibility	10
6. Deliverables of the Project	10
7. Project Constraints and Assumptions	11
7.1 Constraints.....	11
1. User Role Restrictions	11
2. Inventory Management Constraints	11

3. Purchase Order Constraints	11
4. POS and Invoice Constraints	11
5. Domain-Specific Constraints	11
7.2 Project Assumptions	11
1. Data Availability	11
2. User Resources	11
3. Business Operations	11
4. Security and Privacy	12
5. System Usage	12
8. Requirements	12
8.1 Functional Requirements	12
8.1.1 Authentication and Role Management	12
8.1.2 POS and Invoice Management	12
8.1.3 Inventory Management	12
8.1.4 Customer Management	12
8.1.6 Employee and Task Management	12
8.1.7 Expense Management	13
8.1.8 Reports and Analytics	13
8.1.9 Notification Management	13
8.2 Use Case Diagram	13
Use Case Description	14
8.3 Quality Attribute Requirements	15
8.2.1 Security	15
8.2.2 Usability	15
8.2.3 Performance	15
8.2.4 Reliability	15
8.2.5 Maintainability	15
8.2.6 Scalability	15
8.2.7 Data Integrity	15
8.2.8 Compatibility	15
8.4 User Roles	16
1. Business Owner	16
2. Employee	16
3. Customer	17

4. System Administrator	17
5. Supplier.....	18
9. Technologies to be Used.....	19
Frontend.....	19
Backend	19
Database.....	19
Development Tools	19
Development Methodology	19
10. Project timeline.....	19
11. Declaration.....	20

1. Introduction to Project

Problem Statement

Small and medium-scale businesses (SMEs) in Sri Lanka often rely on manual record keeping, spreadsheets, or multiple disconnected software systems to manage their daily operations. As businesses grow, these approaches lead to difficulties in maintaining accurate inventory records, tracking sales and purchases, managing employees, monitoring expenses, and generating reliable business reports. Existing ERP solutions available in the market are often expensive, overly complex, or contain features that are not aligned with the operational requirements of local SMEs. As a result, many businesses continue to operate using fragmented systems, reducing efficiency and limiting their ability to make informed business decisions.

Proposed Solution

To address these challenges, this project proposes the design and development of a modular web-based ERP platform for Sri Lankan SMEs. The system will provide a centralized platform that integrates core business functions including Authentication and Role Management, POS Management, Invoice Management, Inventory Management, Customer Relationship Management (CRM), Supplier Management, Purchase Order Management, Employee Management, Expense Management, Reports and Analytics, and Task Management.

The platform will also support domain-specific operational requirements for selected business sectors such as retail stores, grocery businesses, pharmacies, and hardware stores through specialized features including expiry tracking, weight-based product handling, and unit conversion support. The system will implement role-based access control to support multiple user categories while applying software engineering principles such as modular architecture, database design, system analysis, and web application development using PHP, JavaScript, CSS, and SQL.

2. Project Goal

The goal of this project is to design and develop a modular web-based ERP platform for Sri Lankan SMEs to centralize and streamline business operations through an integrated and customizable system. The platform aims to improve operational efficiency, inventory handling, transaction management, reporting capabilities, and user accessibility while supporting domain-specific operational requirements for selected business sectors including retail stores, grocery businesses, pharmacies, and hardware stores.

3. Scope of the Project

The proposed project focuses on the design and development of a modular web-based ERP platform for Sri Lankan SMEs. The system aims to centralize business operations through integrated management modules while supporting domain-specific operational requirements for selected business sectors.

3.1 In Scope

1. Authentication and Role Management

- User Authentication System
- Role-Based Access Control
 - Business Owner, Employee, Customer, Supplier, and System Administrator roles.
 - Access permissions based on user responsibilities.

2. Point of Sale (POS) and Invoice Management

- POS Operations
 - Product search and billing.
 - Sales transaction processing.
- Invoice Management
 - Invoice generation and storage.
 - Invoice viewing and transaction history management.

3. Inventory and Purchase Management

- Inventory Management
 - Product registration and categorization.
 - Stock level monitoring and inventory updates, Low-stock notifications.
- Purchase Order Management
 - Purchase order creation and tracking.
 - Goods receiving and inventory updates.

4. Customer and Supplier Management

- Customer Relationship Management (CRM)
 - Customer profile management.
 - Customer purchase history tracking.
- Customer Portal
 - View invoices.
 - View purchase history.

- Supplier Management
 - Supplier profile management.
 - Purchase order viewing.
 - Delivery confirmation management.

5. Employee and Task Management

- Employee Management
 - Employee profile management.
 - Employee role assignment.
- Task Management
 - Task creation and assignment.
 - Task progress tracking.

6. Expense Management

- Expense Recording and Monitoring
 - Business expense registration.
 - Expense categorization and tracking.

7. Reporting and Business Insights

- Sales reports.
- Inventory reports.
- Expense reports.

8. Domain-Specific Extensions

- Retail Store Extensions
 - Discount management.
 - Product return management.
- Grocery Business Extensions
 - Weight-based product handling.
 - Batch tracking.
- Pharmacy Extensions
 - Batch management.
 - Expiry date tracking and alerts.
- Hardware Store Extensions
 - Unit conversion management.
 - Support for Piece, Box, Meter, and Kilogram measurements.

3.2 Out of Scope

1. Mobile Application Development

- Native mobile applications for Android and iOS platforms.

2. Advanced Financial Management

- General Ledger Management.
- Payroll Processing.
- Tax Calculation and Compliance Reporting.
- Financial Auditing Modules.

3. Third-Party System Integrations

- External ERP integrations.
- Banking system integrations.
- Third-party accounting software integrations.

4. E-Commerce and Online Services

- E-commerce website integration.
- Online payment gateway integration.
- Online marketplace integrations.

5. Advanced Technologies

- AI-based forecasting and predictive analytics.
- Machine learning-based recommendation systems.
- Real-time IoT device integration.

6. Additional Domain Extensions

- Development of domain-specific extensions beyond Retail Stores, Grocery Businesses, Pharmacies, and Hardware Stores.

4. Objectives of the Project

- To design and develop a modular ERP platform for Sri Lankan SMEs.
- To centralize business operations through a single integrated system.
- To implement role-based access control for different categories of users.
- To develop core ERP modules including POS, inventory, customer, supplier, employee, invoice, purchase order, expense, reporting, and task management.
- To provide domain-specific operational support for retail stores, grocery businesses, pharmacies, and hardware stores.
- To design and implement a secure and efficient relational database for business data management.
- To apply software engineering principles, system analysis techniques, database design concepts, and web development technologies throughout the project lifecycle.

5. Project Feasibility

5.1 Technical Feasibility

The project will be developed using PHP, JavaScript, CSS, HTML, and MySQL technologies, which are well-established and suitable for developing web-based ERP systems. The development team possesses the required technical knowledge and resources to implement the proposed solution.

5.2 Schedule Feasibility

The project scope has been carefully defined to align with the allocated project duration. An Agile incremental development approach will be adopted to ensure timely completion of each module and milestone.

5.3 Resource Feasibility

The required software tools, development environments, and internet access are readily available to the project team. No specialized or costly infrastructure is required for development and testing.

5.4 Economic Feasibility

The project will utilize open-source technologies and freely available development tools, minimizing development costs and making the project economically feasible.

5.5 Ethical and Legal Feasibility

The system will ensure proper handling of user and business information through authentication and access control mechanisms. The project will comply with applicable data privacy and software development ethics while avoiding unauthorized use of proprietary software or data.

6. Deliverables of the Project

The following deliverables will be provided upon project completion:

- A fully functional modular web-based ERP platform.
- Source code of the developed system.
- Database design and implementation scripts.
- System documentation.
- User manual and installation guide.
- Software testing and evaluation reports.
- Project presentation materials and final project report.

7. Project Constraints and Assumptions

7.1 Constraints

1. User Role Restrictions

- Business Owners can manage business settings, employees, branches, and system configurations.
- Employees can only access functions assigned through role-based permissions.
- Customers can only view their invoices, purchase history, and loyalty information.
- Suppliers can only access purchase orders and delivery-related information assigned to them.
- System Administrators can manage users, roles, and system-level configurations.

2. Inventory Management Constraints

- Inventory quantities cannot become negative through sales or stock adjustments.
- Stock movements must be recorded and linked to an authorized user.
- Low-stock notifications will be generated when product quantities reach predefined thresholds.

3. Purchase Order Constraints

- Purchase orders can only be created for registered suppliers.
- Received quantities cannot exceed the quantities specified in the purchase order.
- Completed purchase orders cannot be modified without appropriate authorization.

4. POS and Invoice Constraints

- Every completed sale must generate an invoice record.
- Product sales cannot be processed when sufficient stock is unavailable.
- Returned items must be linked to an existing invoice.

5. Domain-Specific Constraints

- Pharmacy products must contain valid expiry dates and batch information.
- Expired products cannot be sold through the POS module.

7.2 Project Assumptions

1. Data Availability

- All products, customers, suppliers, and employees are registered within the system before operational use.
- Business data entered into the system is accurate and maintained by authorized users.

2. User Resources

- Users have basic computer literacy and access to a modern web browser with internet connectivity.

3. Business Operations

- Businesses operate according to standard inventory, sales, and procurement processes supported by the system.

- Authorized personnel are responsible for maintaining inventory and transactional records.
- 4. **Security and Privacy**
 - User credentials are kept confidential by the respective users.
 - Sensitive business information is accessible only to authorized users according to their assigned roles.
- 5. **System Usage**
 - The selected business domains (Retail, Grocery, Pharmacy, and Hardware) represent the primary operational requirements addressed by the platform.
 - Future modules and business domains may be incorporated through the modular architecture of the system.

8. Requirements

The proposed ERP platform shall provide integrated business management capabilities through a modular architecture while supporting domain-specific operational requirements for selected SME sectors.

8.1 Functional Requirements

8.1.1 Authentication and Role Management

- User registration, login, and logout.
- Role-based access control and permission management.

8.1.2 POS and Invoice Management

- Process sales transactions.
- Generate and manage invoices.
- Handle discounts and returns.

8.1.3 Inventory Management

- Manage products and stock levels.
- Track inventory movements.
- Support batch tracking and expiry management where applicable.
- Support unit conversions and weight-based products where applicable.

8.1.4 Customer Management

- Maintain customer profiles.
- View purchase history.

8.1.5 Supplier and Purchase Order Management

- Manage supplier records.
- Create and manage purchase orders.
- Record goods received from suppliers.

8.1.6 Employee and Task Management

- Manage employee information.

- Assign and track tasks.
- Monitor task completion status.

8.1.7 Expense Management

- Record and categorize business expenses.
- Generate expense summaries.

8.1.8 Reports and Analytics

- Generate sales, inventory, purchase, expense, and employee activity reports.

8.1.9 Notification Management

- The system shall generate notifications for low stock levels, overdue invoices, and important system events.
- Users shall be able to view notifications relevant to their role.

8.2 Use Case Diagram

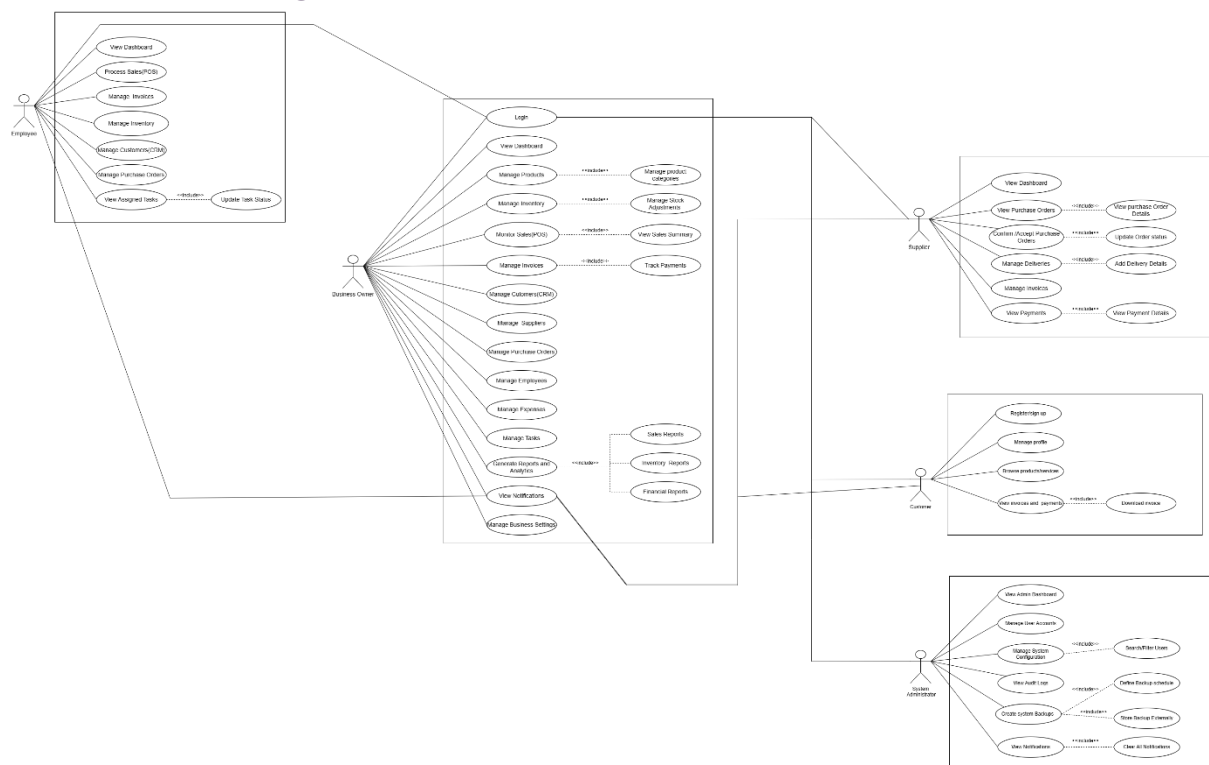


Figure 8.2 : Complete Use Case Diagram

Link :

https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=blank&layers=1&nav=1&dark=auto#Uhttps%3A%2Fdrive.google.com%2Fuc%3Fid%3D1rULHi-bo40a6ib_tJbQ1aXVURQg4G9Wvy%26export%3Ddownload

Use Case Description

Use Case	Primary Actor(s)	Description
Login	All Users	Allows authorized users to securely access the ERP system according to their assigned roles and permissions.
Manage Products	Business Owner, Employee	Enables users to add, update, categorize, and view products maintained within the business inventory.
Manage Inventory	Business Owner, Employee	Allows monitoring of stock levels, stock movements, adjustments, expiry tracking, and inventory availability across business operations.
Process Sales (POS)	Employee	Enables employees to process customer purchases, calculate totals, generate receipts, and update inventory automatically.
Manage Invoices	Employee, Business Owner	Allows creation, viewing, updating, and tracking of invoices and customer payment records.
Manage Customers (CRM)	Business Owner, Employee	Enables management of customer information, purchase history, and customer interactions.
Manage Suppliers	Business Owner	Allows maintenance of supplier records, supplier communication details, and supply-related information.
Manage Purchase Orders	Business Owner, Supplier	Enables creation, tracking, and management of purchase orders for replenishing business inventory.
Manage Employees	Business Owner	Allows the management of employee information, responsibilities, and system access privileges.
Manage Expenses	Business Owner	Enables recording and monitoring of operational expenses for financial tracking and analysis.
Generate Reports & Analytics	Business Owner	Provides business performance reports, sales summaries, inventory reports, and analytical insights for decision-making.
Manage Tasks	Business Owner, Employee	Allows creation, assignment, monitoring, and completion of operational tasks within the organization.
Manage Users & Roles	System Administrator	Enables administration of user accounts, role assignments, permissions, and access control policies.

View Notifications	All Users	Allows users to receive and view system-generated alerts, reminders, and important operational updates.
View Dashboard	Business Owner, Employee	Displays key business information, statistics, and performance indicators through a centralized dashboard interface.

Table 1: Use Case Description

8.3 Quality Attribute Requirements

8.2.1 Security

- User passwords are encrypted before storing in the database. Each user can only access features allowed by their role. All login and data change activities are recorded in a log.

8.2.2 Usability

- Role-based dashboards show only relevant modules per user. Responsive CSS layout supports desktop, tablet, and mobile. Inline JavaScript validation provides real-time form feedback. Sinhala, Tamil, and English receipt toggle supported.

8.2.3 Performance

- Efficient database design and optimized queries will be used to ensure acceptable response times.

8.2.4 Reliability

- Data validation and error-handling mechanisms will be implemented to maintain system stability.

8.2.5 Maintainability

- Modular system architecture and proper documentation will support future maintenance and enhancements.

8.2.6 Scalability

- The modular design will allow additional business domains and modules to be incorporated in future versions.

8.2.7 Data Integrity

- The system shall ensure that business transactions, inventory records, invoices, and customer information remain accurate and consistent throughout all operations by enforcing database constraints and validation rules.

8.2.8 Compatibility

- The system shall be accessible through modern web browsers and support various screen sizes including desktops, tablets, and mobile devices through responsive web design.

8.4 User Roles

1. Business Owner



Figure 8.4.1: Business Owner Use Case Diagram.

2. Employee

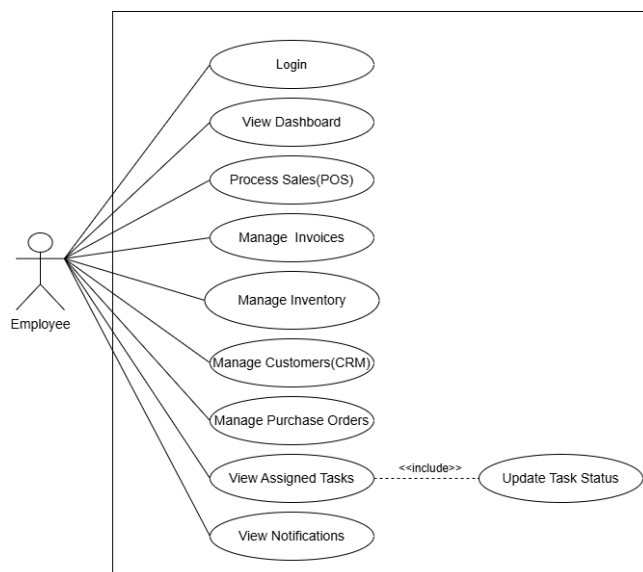


Figure 8.4.2: Employee Use Case Diagram.

3. Customer

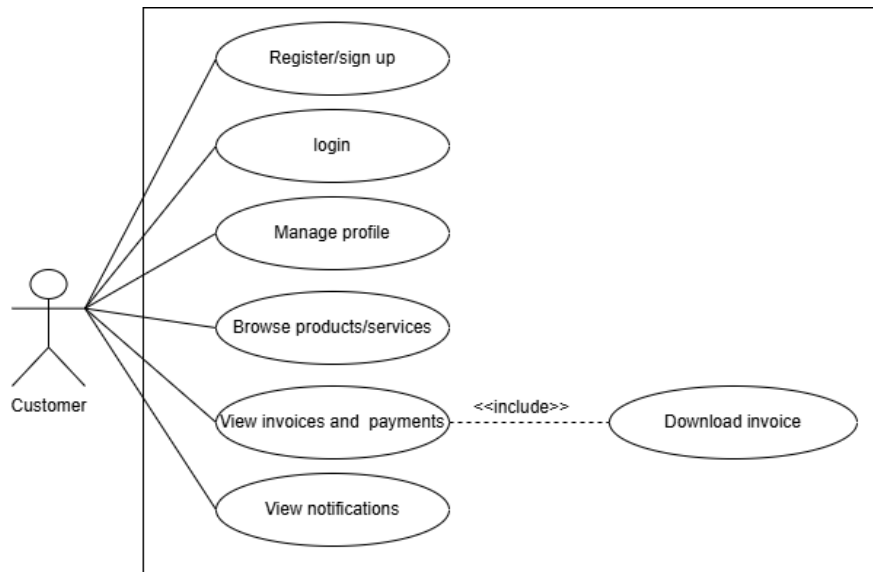


Figure 8.4.3: Customer Use Case Diagram.

4. System Administrator

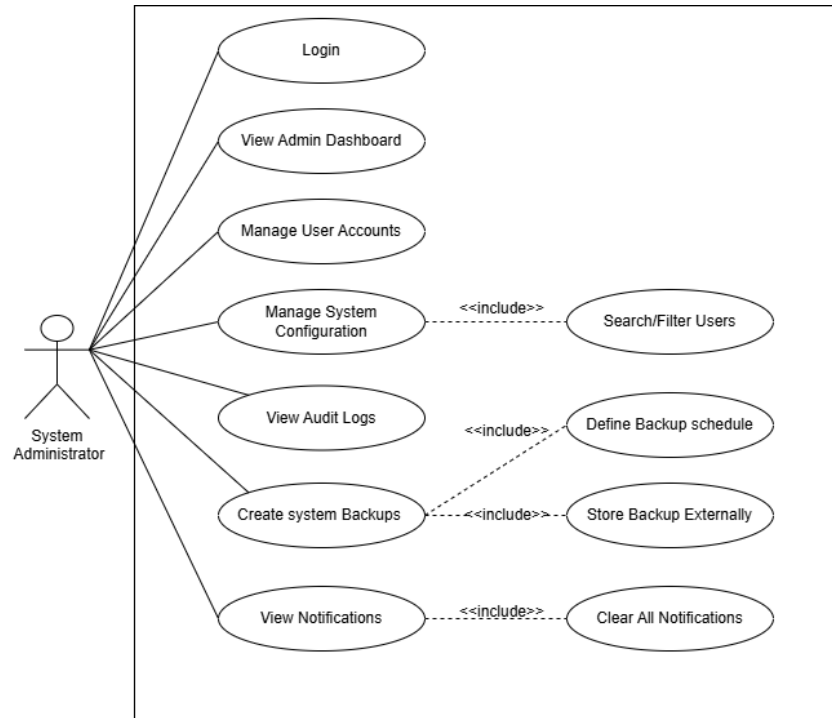


Figure 8.4.4: System Administrator Use Case Diagram.

5. Supplier

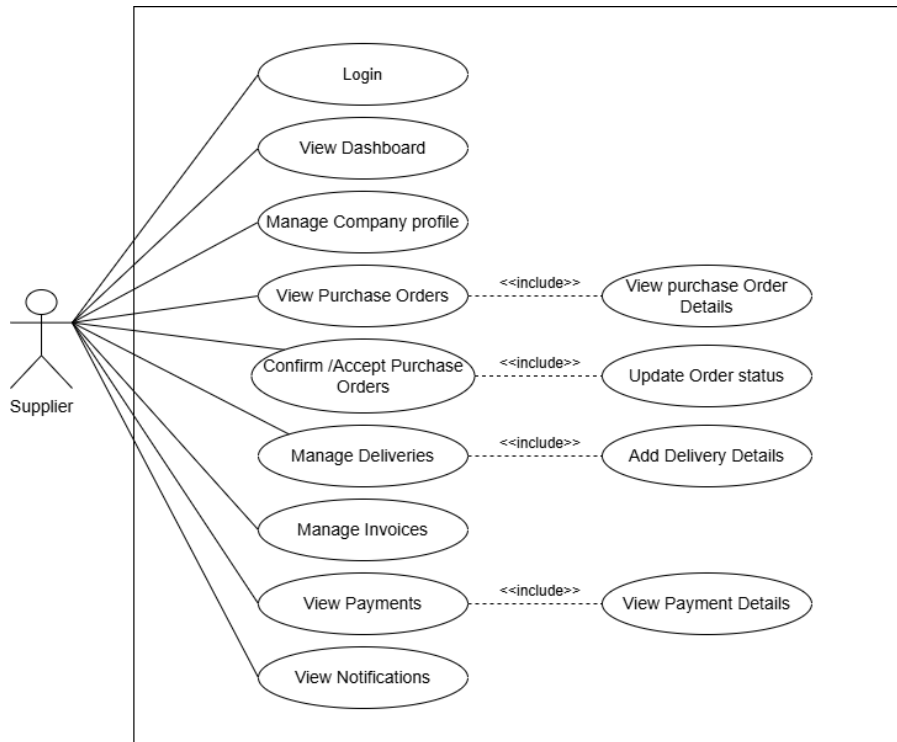


Figure 8.4.5: Supplier Use Case Diagram.

Figure No.	Figure Title	Brief Explanation
Figure 8.2	Complete Use Case Diagram	Illustrates the overall interactions between the main user roles and the core functional modules of the SmartERP system.
Figure 8.4.1	Business Owner Use Case Diagram	Shows the functions accessible to the Business Owner, including inventory, employee, supplier, expense, and report management.
Figure 8.4.2	Employee Use Case Diagram	Illustrates employee activities such as POS operations, invoice handling, inventory updates, and task management.
Figure 8.4.3	Customer Use Case Diagram	Represents customer interactions including viewing invoices and purchase history.
Figure 8.4.4	System Administrator Use Case Diagram	Shows administrative functions such as managing users, roles, permissions, and system configurations.
Figure 8.4.5	Supplier Use Case Diagram	Illustrates supplier-related activities including viewing purchase orders and delivery information.

Table 2 : User Role Use Cases Explanation

9. Technologies to be Used

Frontend

- HTML
- CSS
- JavaScript

Backend

- PHP

Database

- MySQL

Development Tools

- Visual Studio Code
- Git and GitHub
- XAMPP

Development Methodology

- Agile Incremental Development Model

10. Project timeline

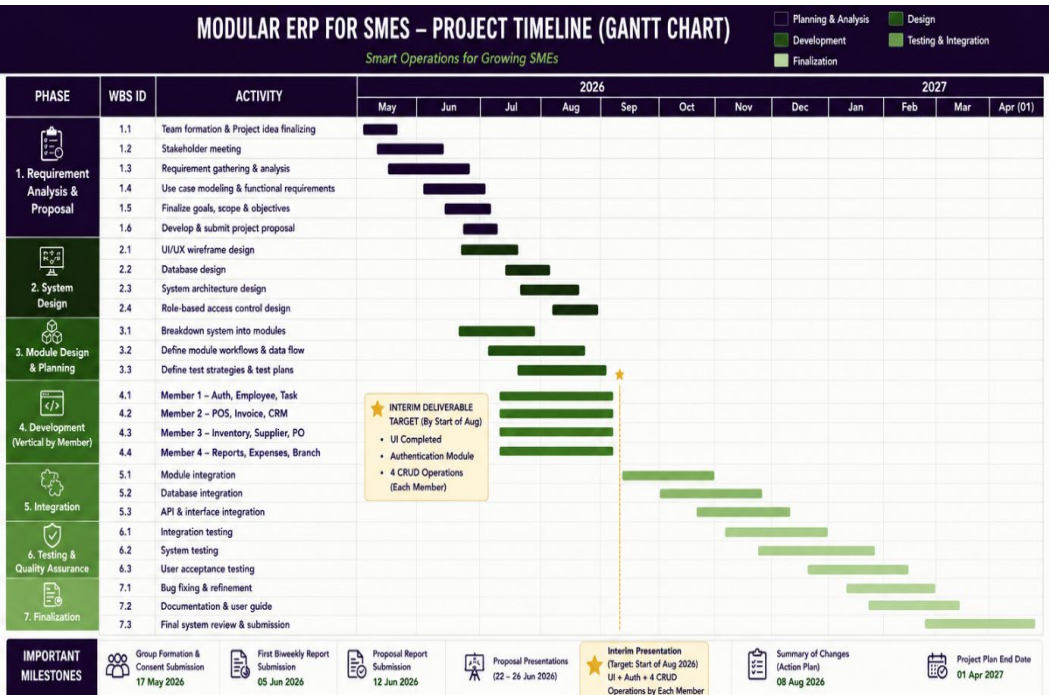


Figure 10 : Project Timeline

Chart Link :

<https://docs.google.com/spreadsheets/d/1Eu3zCSapD9F5xuDXpaGUtVJbfsPSexmYyYraVULVMA/edit?usp=sharing>

11. Declaration

We as members of the project titled SmartERP, Certify that we will carry out this project according to the guidelines provided by the coordinators and supervisors of the course as well as we will not incorporate, without acknowledgment, any material previously submitted for a degree or diploma in any university. To the best of our knowledge and brief, the project work will not contain any material previously published or written by another person or ourselves except where due reference is made in the text of appropriate places

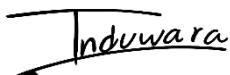
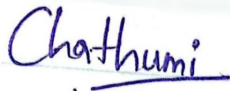
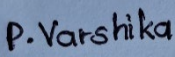
Index Number	Name of the student	Signature
24021067	R. A. S. Thivanka	
24021059	K. I. U. Thisera	
24020451	L.A.C.R. JAYAMALI	
24020771	V. PAVALARAJ	

Table 3 : Team Members